
Lecture Notes

on Geometric and Arithmetic Sequences, Linear Difference Equation

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1 Geometric Sequences

1.1 Definition

A **geometric sequence** is a sequence in which each term is obtained by multiplying the previous term by a fixed real number r , called the **common ratio**.

Formally, a geometric sequence is defined by

$$x_{n+1} = rx_n, \quad n = 0, 1, 2, \dots \quad (1)$$

with initial value $x_0 \in \mathbb{R}$.

1.2 General Formula

From (1), we compute successive terms:

$$\begin{aligned} x_1 &= rx_0, \\ x_2 &= rx_1 = r(rx_0) = r^2x_0, \\ x_3 &= rx_2 = r(r^2x_0) = r^3x_0, \\ &\vdots \\ x_n &= r^n x_0. \end{aligned}$$

Thus, the **general solution** of a geometric sequence is

$$x_n = r^n x_0, \quad n \geq 0.$$

1.3 Examples

- If $r = \frac{1}{2}$ and $x_0 = 1$, then

$$x_n = \left(\frac{1}{2}\right)^n, \quad \text{sequence: } 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$$

- If $r = 2$ and $x_0 = 1$, then

$$x_n = 2^n, \quad \text{sequence: } 1, 2, 4, 8, 16, \dots$$

1.4 Behavior of a Geometric Sequence

If $x_0 \neq 0$, the behavior of $x_n = r^n x_0$ depends on the common ratio r :

- If $0 < r < 1$, then $x_n \rightarrow 0$ as $n \rightarrow \infty$ (decay to zero).

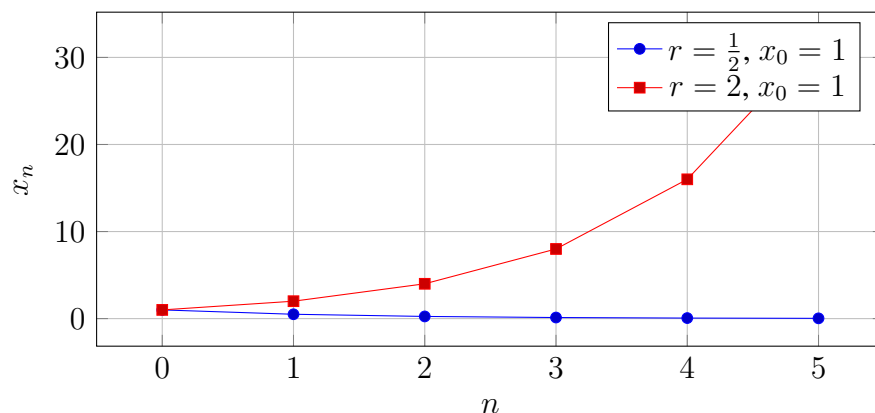
- If $r > 1$, then $x_n \rightarrow \infty$ as $n \rightarrow \infty$ (exponential growth).

- If $-1 < r < 0$, then $x_n \rightarrow 0$ but oscillates in sign.
- If $r \leq -1$, the sequence oscillates with increasing magnitude and does not converge.

1.5 Graphical Illustration

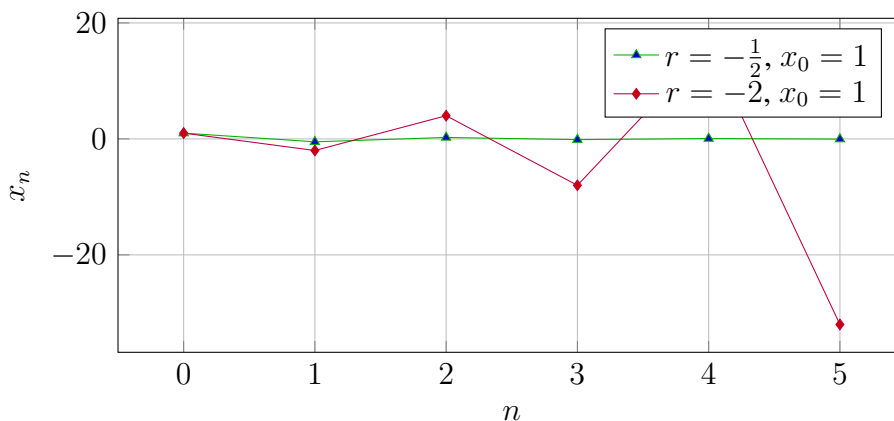
Case 1: $0 < r < 1$ and $r > 1$

Geometric Sequences: Decay and Growth



Case 2: $-1 < r < 0$ and $r \leq -1$

Geometric Sequences: Oscillatory Behavior



✎ **Example:** Let $x_0 = 5$ and $r = \frac{3}{2}$.

1. Write the geometric sequence with its general solution in terms of x_n .
2. Find x_1 , x_2 , and x_3 .
3. Determine whether the sequence converges or diverges.

$$x_n = \frac{3}{2} x_{n-1}, \\ n \geq 1$$

1) Geometric sequence: $x_{n+1} = \frac{3}{2} x_n, n \geq 0$.

General soln: $x_n = \left(\frac{3}{2}\right)^n x_0 = \left(\frac{3}{2}\right)^n 5, n \geq 0$. — (*)

2) (*) $\Rightarrow$

$$x_1 = \left(\frac{3}{2}\right)^1 5 = 7.5$$

$$x_2 = \left(\frac{3}{2}\right)^2 5 = 11.25$$

$$x_3 = \left(\frac{3}{2}\right)^3 5 = 16.875$$

3) Here, $r = \frac{3}{2} > 1$, so the sequence will diverge.

2 Arithmetic Sequences

2.1 Definition

A **arithmetic sequence** is a sequence in which each term is obtained by adding a fixed real number d to the previous term. The constant d is called the **common difference**.

Formally, an arithmetic sequence is defined by

$$x_{n+1} = x_n + d, \quad n = 0, 1, 2, \dots \quad (2)$$

with initial value $x_0 \in \mathbb{R}$.

2.2 General Formula

From (2), we compute successive terms:

$$\begin{aligned}x_1 &= x_0 + d, \\x_2 &= x_1 + d = (x_0 + d) + d = x_0 + 2d, \\x_3 &= x_2 + d = (x_0 + 2d) + d = x_0 + 3d, \\&\vdots \\x_n &= x_0 + nd.\end{aligned}$$

Thus, the **general solution** of an arithmetic sequence is

$$x_n = x_0 + nd, \quad n \geq 0.$$

$$\begin{aligned}x_{n+1} &= x_n + d \\x_n &= x_{n-1} + d, \\n &\geq 1.\end{aligned}$$

2.3 Examples

- If $x_0 = 3$ and $d = 2$, then

$$x_n = 3 + 2n, \quad \text{sequence: } 3, 5, 7, 9, 11, \dots$$

- If $x_0 = 10$ and $d = -3$, then

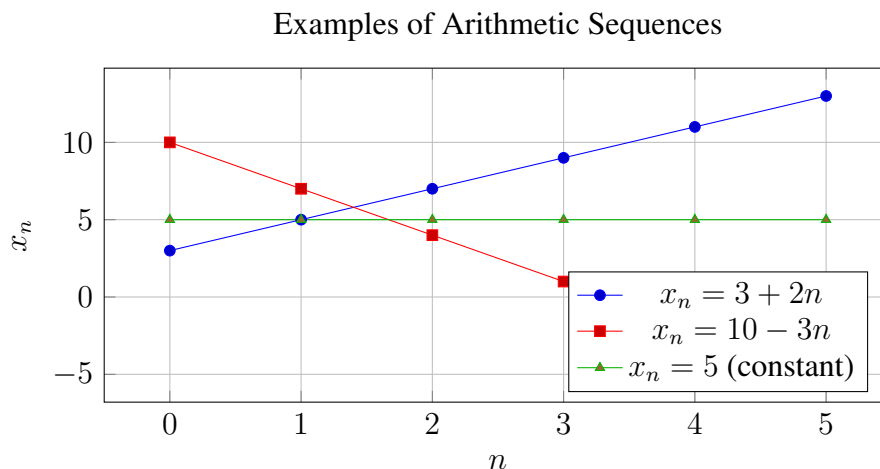
$$x_n = 10 - 3n, \quad \text{sequence: } 10, 7, 4, 1, -2, \dots$$

2.4 Behavior of an Arithmetic Sequence

The long-term behavior depends on the common difference d :

- If $d > 0$, the sequence increases without bound ($x_n \rightarrow +\infty$).
- If $d < 0$, the sequence decreases without bound ($x_n \rightarrow -\infty$).
- If $d = 0$, the sequence is constant ($x_n = x_0$).

2.5 Graphical Illustration



✎ **Example:** Let $x_0 = 4$ and $d = -2$.

1. Write the arithmetic sequence with its general solution in terms of x_n .
2. Find x_1 , x_2 , and x_3 .
3. Determine whether the sequence converges or diverges.

1) Arithmetic sequence:

$$x_{n+1} = x_n + d = x_n - 2, \quad n \geq 0$$

General solnⁿ:

$$x_n = x_0 + nd = 4 - 2n, \quad n \geq 0. \quad \text{---} (*)$$

2) $(*) \Rightarrow x_1 = 4 - 2 = 2$

$$x_2 = 4 - 2 \cdot 2 = 4 - 4 = 0.$$

$$x_3 = 4 - 2 \cdot 3 = 4 - 6 = -2.$$

3) As $d = -2$, the sequence is decreasing and it is a divergent sequence.

3 Linear Difference Equations with Constant Coefficients

3.1 Definition

A first-order difference equation is called **linear** if it can be written in the form

$$x_{n+1} = a_n x_n + b_n,$$

where $\{a_n\}$ and $\{b_n\}$ are sequences of constants.

In this section, we consider the special case where $a_n = a$ and $b_n = b$ are fixed real constants. Thus, the difference equation reduces to

$$x_{n+1} = ax_n + b. \quad (3)$$

Special cases:

- If $a = 1$, then $x_{n+1} = x_n + b$, which is an **arithmetic sequence**.
- If $a \neq 1$ and $b = 0$, then $x_{n+1} = ax_n$, which is a **geometric sequence**.

3.2 General Solution of (3)

Step 1: Homogeneous Part

Consider the corresponding homogeneous equation:

$$x_{n+1} = ax_n.$$

The solution is

$$x_n = a^n C, \quad C \in \mathbb{R}.$$

$C = ?$

Step 2: Particular Solution

Assume a constant solution $x_n = K$. Substituting into (3),

$$K = aK + b \Rightarrow K(1 - a) = b \Rightarrow K = \frac{b}{1 - a}, \quad (a \neq 1).$$

Step 3: General Solution

The general solution of (3) is

$$x_n = a^n C + \frac{b}{1 - a}.$$

General soln of (3)
= hom. part solnⁿ
+ particular solnⁿ.

Step 4: Applying Initial Condition

If x_0 is the given initial condition, then

$$x_0 = a^0 C + \frac{b}{1 - a} \Rightarrow C = x_0 - \frac{b}{1 - a}.$$

Hence, the solution becomes

$$x_n = a^n \left(x_0 - \frac{b}{1 - a} \right) + \frac{b}{1 - a}.$$

$$\Rightarrow x_n = a^n x_0 - \frac{a^n b}{1 - a} + \frac{b}{1 - a} \\ = a^n x_0 + \frac{b(1 - a^n)}{(1 - a)}$$

Alternatively, it can be written as

$$x_n = a^n x_0 + \frac{b(1 - a^n)}{1 - a}.$$

3.3 Long-Term Behavior

If $|a| < 1$, then as $n \rightarrow \infty$, $a^n \rightarrow 0$. Thus,

$\hookrightarrow -1 < a < 1$

$$\lim_{n \rightarrow \infty} x_n = \frac{b}{1 - a}.$$

If $|a| \geq 1$, the sequence typically diverges (depending on x_0 and b).

$\hookrightarrow a \geq 1$ and $a \leq -1$.

Example: (Lake Fish Population Model) Consider a lake fish population whose yearly birth rate is 1.2 and yearly death rate is 0.7. Each year, fishing is allowed until 1200 fish are caught. There are currently 12,230 fish in the lake.

1. Write a difference equation for the lake fish population and find the general solution.
2. How many fish are in the lake after 5 years?
3. If the resource managers of the lake want the population to remain constant each year, what level of harvesting should they allow? (for stability)

1) Let the difference equation be given by

$$x_{n+1} = ax_n + b \quad (*)$$

where, $a = 1 + 1.2 - 0.7 = 1.5$

& $b = -1200$

General solution,

$$x_n = a^n \left(x_0 - \frac{b}{1-a} \right) + \left(\frac{b}{1-a} \right),$$

where $x_0 = 12230$. $n \geq 0$.

$$\begin{aligned} \therefore x_n &= (1.5)^n \left(12230 - \frac{-1200}{1-1.5} \right) + \left(\frac{-1200}{1-1.5} \right) \\ &= (1.5)^n (9830) + 2400. \end{aligned}$$

2) $x_5 = (1.5)^5 (9830) + 2400 \approx 77047$

3) As the population need to remain constant each year, we must have $x_{n+1} = x_n = x_{n+1} = \dots = x_1 = x_0 = \dots$

$x_1 = x_0 \Rightarrow 1.5x_0 + b = x_0$

From equation (*)

For simplicity we will use this

Due birth increase
 $1.2x_n$

Due death decrease
 $0.7x_n$

So, net effect

$$x_n + 1.2x_n - 0.7x_n$$

$$= x_n (1 + 1.2 - 0.7)$$

$$= 1.5x_n$$

$$\Rightarrow (1.5-1)x_0 + b = 0$$

$$\Rightarrow b = -(1.5-1)x_0 = -0.5 \times 12230 = -6115.$$

Thus they should allow 6115 number of fish to harvest.

✍ **Example: (Drug Accumulation in the Body)** A patient takes 15 mg of a drug at the same time each day. If 80% of the drug is removed from the body each day, how much of the drug will remain in the body after a long period of time (i.e., as $n \rightarrow \infty$)?

Here $a = (100 - 80)\% = 20\% = 0.2$.
After a long period of time the amount of the drug
will remain $\left(\frac{b}{1-a}\right) \text{ mg}$
$$= \frac{15}{(1-0.2)} = \frac{15}{0.8} = 18.75 \text{ mg}.$$